Operating Systems Numerical Questions

Process Scheduling, Paging & TLB, Page Replacement Policies

Part A: Process Scheduling (MCQs)

1. Consider three processes with arrival times and burst times:
   P1: Arrival = 0, Burst = 5;
   P2: Arrival = 1, Burst = 3;
   P3: Arrival = 2, Burst = 8.
   Using FCFS scheduling, what is the average waiting time?

   (a) 4.33 ms

   (b) 5.33 ms

   (c) 6.33 ms

   (d) 7.33 ms

2. Consider the same processes as above. Using SJF (non-preemptive), what is the average turnaround time?

   (a) 10.33 ms

   (b) 9.33 ms

   (c) 8.33 ms

   (d) 7.33 ms

3. Four processes P1, P2, P3, P4 have burst times 5, 15, 4, 3 respectively, and arrival times 0. If scheduled with Round Robin (time quantum = 4), what is the average waiting time?

   (a) 8.5 ms

   (b) 10.5 ms

   (c) 12.5 ms

   (d) 14.5 ms

4. Two processes P1 and P2 arrive at time 0 with burst times 6 and 4. Priority scheduling is used with priority(P1)=2, priority(P2)=1 (lower value = higher priority). What is the waiting time of P1?

   (a) 4 ms

   (b) 6 ms

   (c) 8 ms

   (d) 10 ms

5. Three processes with burst times 10, 5, and 8 are scheduled with Round Robin (time quantum = 2). What is the average turnaround time?

   (a) 15.67 ms

   (b) 16.67 ms

   (c) 17.67 ms

   (d) 18.67 ms

Part B: Page Replacement Policies (MCQs)

1. Reference string: 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2 with 3 frames. Using FIFO, how many page faults occur?

   (a) 9

   (b) 10

   (c) 11

   (d) 12

2. With the same reference string and 3 frames, using LRU, how many page faults occur?

   (a) 9

   (b) 10

   (c) 11

   (d) 12

3. Consider Optimal replacement policy on the same string and 3 frames. Number of page faults?

   (a) 7

   (b) 8

   (c) 9

   (d) 10

4. Reference string: 1,2,3,4,1,2,5,1,2,3,4,5 with 3 frames. Using FIFO, number of page faults?

   (a) 8

   (b) 9

   (c) 10

   (d) 11

5. Using LRU on the same string and 3 frames, number of page faults?

   (a) 9

(b) 10

(c) 11

(d) 12

Part C: Paging and TLB (Numerical Questions)

1. A system has 32-bit logical address space with page size = 4KB. Physical memory is 30-bit. Find size of page table in bytes (assuming 4 bytes per entry).

2. Logical Address = 16 bits, Page Size = 1 KB, Physical Address = 14 bits. Find number of entries in the page table.

3. TLB hit ratio = 90%, TLB access time = 10 ns, Memory access time = 100 ns. Find effective memory access time.

4. A 32-bit system has 1 GB physical memory and 4 KB page size. Find number of frames.

5. Logical Address = 28 bits, Page Size = 2 KB, Physical Address = 24 bits. Find page table size (in bytes) assuming 4 bytes per entry.

6. TLB access time = 20 ns, Memory access time = 100 ns, Hit ratio = 80%. Find effective memory access time.

7. If a process generates 200 memory references, TLB hit ratio = 85%, memory access = 50 ns, TLB access = 5 ns. Find average time.

8. Logical address = 32 bits, Page size = 8 KB. How many entries are in the page table?

9. Physical memory = 256 MB, Page size = 4 KB. How many frames are there?

10. Given TLB hit ratio = 95%, TLB = 10 ns, Memory = 80 ns. Compute effective access time.